

Symmetric Matrices:

$$A = A^T$$

Eigen-values: $\mathbb{R}$

Eigen-vectors are perpendicular/orthogonal

↳ Can be made unit vectors to get ortho-normal vectors

$$A = S \Lambda S^{-1}$$

$$= Q \Lambda Q^{-1}$$

↓
ortho-normal

$$= Q \Lambda Q^T \rightarrow \text{Principle Axis / Spectral theorem}$$

Why real eigen-values?

$$Ax = \lambda x \xrightarrow{\text{always}} A\bar{x} = \bar{\lambda} \bar{x} \text{ (conjugate)}$$

$$\bar{x}^T A x = \lambda \bar{x}^T x \quad \text{--- (1)} \quad \bar{x}^T A^T = \bar{x}^T \bar{\lambda}$$

$$\bar{x}^T A = \bar{x}^T \bar{\lambda}$$

$$\bar{x}^T A x = \bar{x}^T \bar{\lambda} x \quad \text{--- (2)}$$

From (1) & (2);

$$\lambda \boxed{\bar{x}^T x} = \bar{\lambda} \boxed{\bar{x}^T x} \quad \swarrow \text{len}^2$$

$$\lambda = \bar{\lambda}$$

So, λ is $\mathbb{R}$

$$= [\bar{x}_1 \ \bar{x}_2 \ \dots \ \bar{x}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \bar{x}_1 x_1 + \bar{x}_2 x_2 + \dots$$

$$= (a - ib)(a + ib) + \dots$$

$$= a^2 + \cancel{abi} - \cancel{abi} + b^2$$

$$= a^2 + b^2$$

$$A = Q \Lambda Q^T$$

$$= \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix}$$

$$= \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \dots$$

Every symmetric matrix is a combination of $\perp$ projection matrices

↳ Signs of pivot are same as sign of eigen-values.

+ve pivots = # +ve eigen-values

Positive Definite Matrix:

↳ It is symmetric

↳ All +ve eigen-values

↳ All pivots are +ve

↳ All sub-determinants are +ve

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

① $\lambda_1, \lambda_2 > 0$

② $a > 0, ac - b^2 > 0$

③ Pivots $a > 0$

④ $x^T A x > 0$
except at $x=0$

Example,

$$A = \begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix} \quad 2(18) - 6(6) = 0 \quad A = \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix}$$

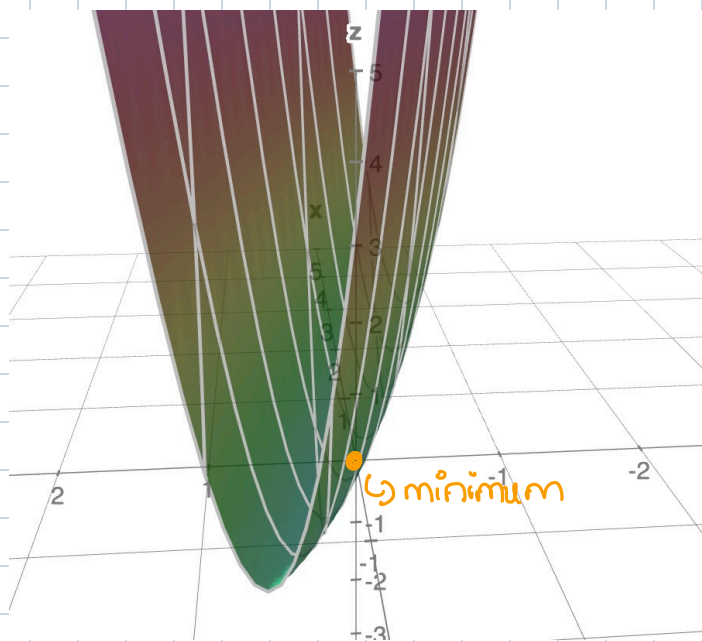
↳ Pos semi-definite

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 + 6x_2 \\ 6x_1 + 20x_2 \end{bmatrix}$$

$$= 2x_1^2 + 12x_1x_2 + 20x_2^2$$

↳ Quadratic form
↳ > 0 for all x_1, x_2



↗ pivots

$$= 2(x + 3y)^2 + 2y^2$$

↖ pivots

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \sim \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$R_2 - 3R_1 \rightarrow R_2$

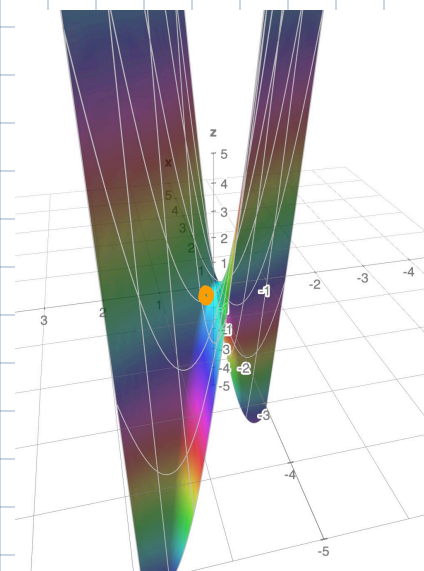
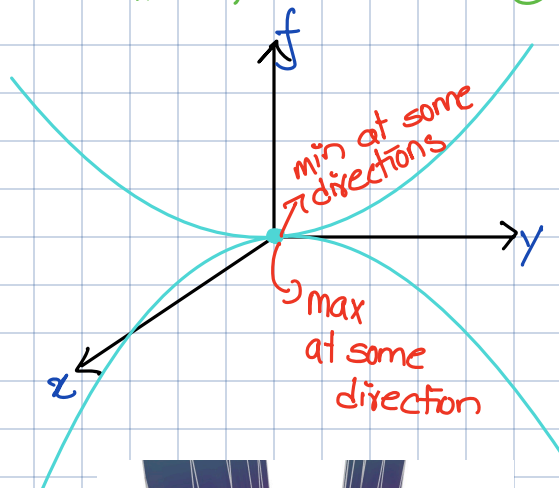
$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 + 6x_2 \\ 6x_1 + 7x_2 \end{bmatrix}$$

$$f(x,y) = 2x_1^2 + 12x_1x_2 + 7x_2^2$$

$$\hookrightarrow x_1 = 1, x_2 = -1 < 0$$

↳ up in some directions
↳ down in others
↳ saddle



$$f(x_1, x_2, \dots, x_n)$$

Min $\sim$ Matrix of 2nd derivative is pos definite

$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 \\ 12 & 14 \end{bmatrix}$$

$\hookrightarrow$ This matrix is NOT positive definite. So, we have a saddle point.

$$f_{x_1} = 4x_1 + 12x_2$$

$$f_{x_2} = 12x_1 + 14x_2$$

$$f_{x_1 x_2} = 12$$

$$f_{x_2 x_1} = 12$$

$$f_{x_1 x_1} = 4$$

$$f_{x_2 x_2} = 14$$

A is positive definite, then is A^{-1} as well?

$$\text{Eigen-values of } A^{-1} = \frac{1}{\text{Eigen-values of } A}$$

If A & B are positive definite, then A+B

$$x^T(A+B)x$$

Yes!

$A_{m \times n} \rightarrow \text{Rank} = n$ then

$[A^T A] \rightarrow$ square & symmetric, pos definite

$$x^T A^T A x$$

$$(Ax)^T (Ax)$$

$$\|A_x\|^2 \geq 0$$